MINI PROJECT REVIEW

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Comparison of Machine Learning Techniques for Artist Identification

Logistic Regression, SVM, CNN and a Random Forest extension, evaluated on 7,462 paintings from the Painters by Numbers dataset.

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GITHUB REPOSITORY

github.com/sheetalbevinakatti/Artist_Identification

SECTION 01

Problem Statement & Dataset

Multiclass classification: identify the artist of a painting from its image.

7,462
paintings selected
Painters by Numbers

15
artist classes
multiclass task

256 × 256
RGB image size
input to the CNN

OUR TRAIN / VALIDATION / TEST SPLIT

5,967	**Training**	≈ 80% of images
746	**Validation**	≈ 10% of images
749	**Test**	≈ 10% of images

Note: Our split differs slightly from the reference paper's split, so results are not directly one-to-one comparable.

SECTION 02

Existing Approaches & Our Extension

Handcrafted features vs. learned features, plus one add-on model.

BASELINES

Logistic Regression & SVM

- Handcrafted features: GIST, color histogram, Hu moments
- Our GIST is a Gabor-filter-based approximation
- SVM tuning is simplified; the paper's full search is not reproduced

DEEP LEARNING

Convolutional Neural Network

- Learns features directly from 256 × 256 RGB images
- Trained in Kaggle
- The saved model is reused by our Streamlit demo

OUR EXTENSION

Random Forest

- Not evaluated in the reference paper
- 200 trees on GIST + color histogram features
- Tests whether a tree ensemble beats the SVM on the same handcrafted features

Repository scripts (src/):
`extract_features.py · train_logistic_regression.py · train_svm.py · train_cnn.py · train_random_forest.py`

SECTION 03

Our Results vs. the Reference Paper

Test accuracy: paper-reported values next to our own measured values (partial reproduction).

Model	Paper	Ours	Diff. (pts)
Logistic Regression	61.2%	**57.54%**	−3.66
SVM · GIST + Hu moments	68.1%	**47.40%**	−20.70
SVM · GIST + color histogram	67.3%	**55.94%**	−11.36
SVM · GIST + color hist. + Hu moments	61.2%	**55.94%**	−5.26
CNN	**74.7%**	**65.55%**	−9.15
Random Forest (our extension)	*not reported*	**56.48%**	n/a

WHY THE GAPS

- Split differs slightly from the paper
- GIST is a Gabor-filter-based approximation
- SVM tuning is simplified

Partial reproduction: paper values and our values are reported separately; we do not claim an exact reproduction. CNN has the highest test accuracy in both.

SECTION 04

Extension Model: Random Forest

An add-on model that the reference paper did not evaluate.

Setup & rationale

- Extension: not one of the reference paper's models
- 200 decision trees
- Input: GIST + color histogram features
- Question: does a tree ensemble generalize better than a single-boundary SVM on the same features?

Measured accuracy

Training	Validation	Test
100.00%	56.17%	56.48%

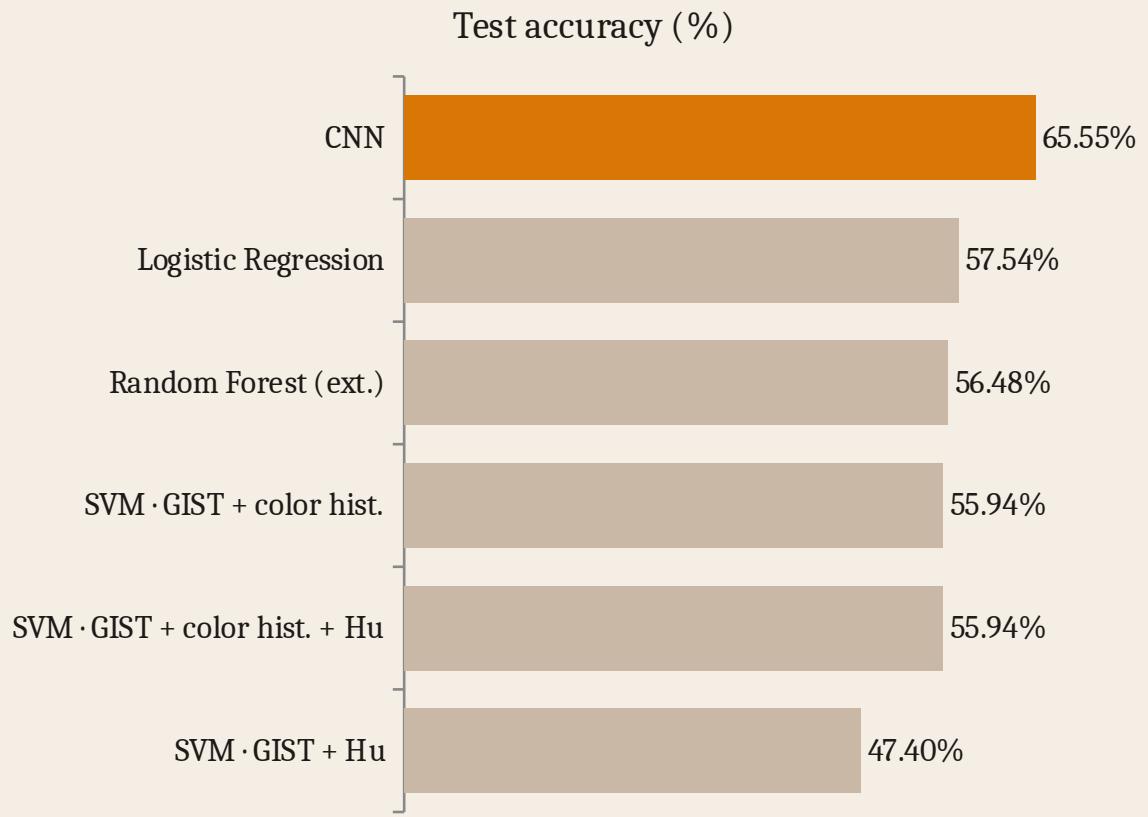
100% training vs. 56.48% test suggests overfitting and poor generalization on these handcrafted features, not robustness.

In context: 56.48% is slightly above our best SVM (55.94%) and well below our CNN (65.55%).

SECTION 05

Comparison & Key Findings

Our measured test accuracy for every model we evaluated.



CNN was best in our experiments

65.55% test accuracy, about 8 to 18 points above the other models.

Gaps between training and test

RF: 100.00% train vs. 56.48% test. CNN: 99.87% train vs. 65.68% final validation (best 66.22% at epoch 26).

Differences from the paper

Our numbers are lower, likely due to the split and implementation choices (GIST approximation, simplified SVM).

SECTION 06

Working Streamlit Demonstration

Local web app that runs the saved CNN on a painting.

HOW IT WORKS

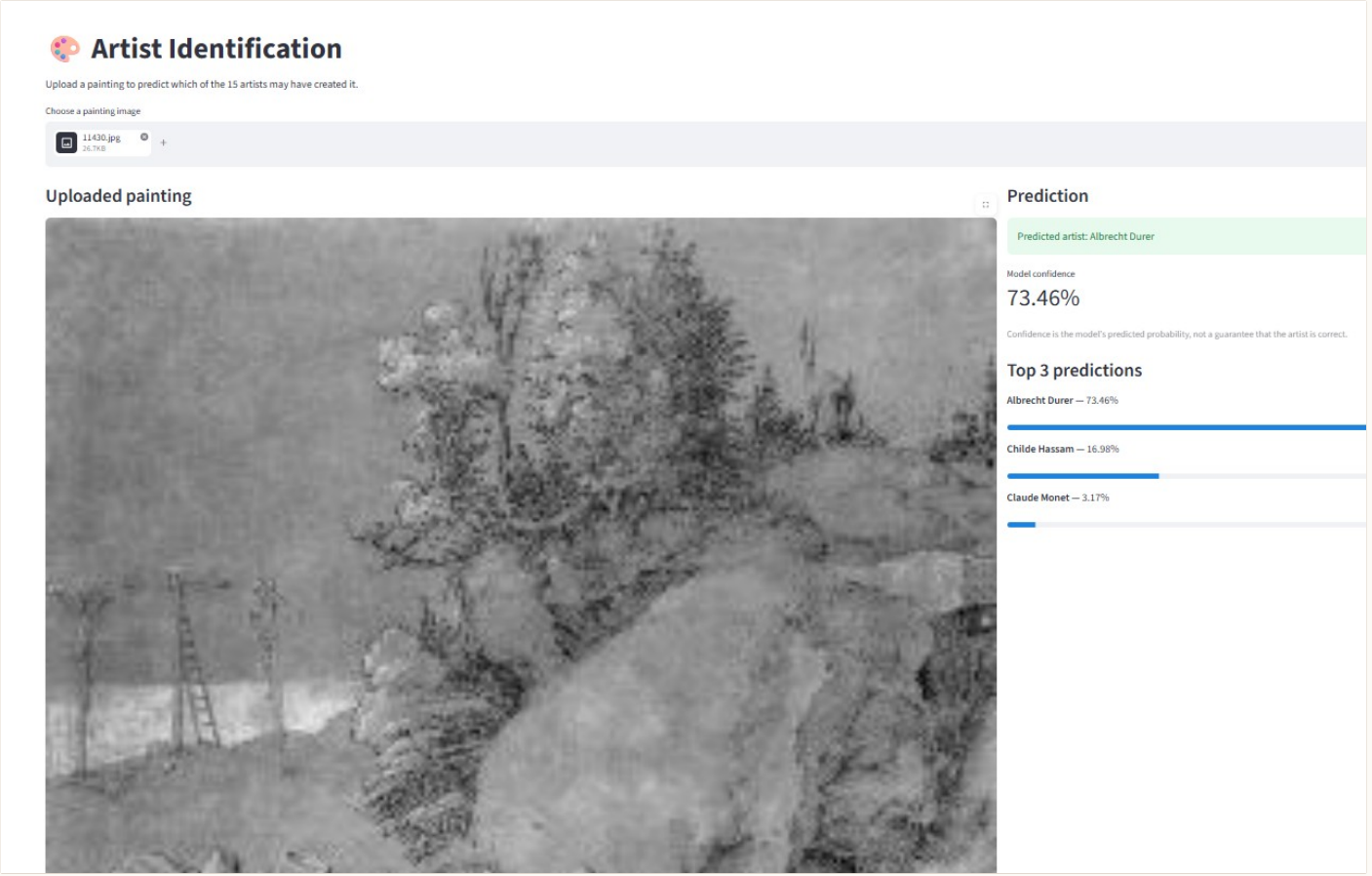
- Local Streamlit app loads the saved CNN (trained in Kaggle)
- Returns the predicted artist, confidence and top-3 scores
- Inference only: the app does not retrain the model

TEST CASE (SCREENSHOT)

11430.jpg · test split · label: Albrecht Durer

Screenshot shows: Albrecht Durer, 73.46%

One example illustrates the workflow; it does not show overall accuracy.



SUMMARY

Conclusion

Compared three model approaches from the reference study, evaluated Random Forest as an extension, and built a working CNN prediction demo.

65.55%

CNN: best test accuracy in our experiments

56.48%

Random Forest extension: test accuracy (100.00% train)

Partial

reproduction of the reference study, plus one extension

REPOSITORY **github.com/sheetalbevinakatti/Artist_Identification**